The value of *mass* information

Search in the Google era

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A broad class of economic decision problems involves sequentially searching over a collection of uncertain research leads. Existing theory informs the efficient search over this pool using available information. But in today's information age, the researcher has the opportunity to collect additional information *en masse*, e.g. with a search engine or scientific framework, that will benefit the search. We derive the value of this type of mass information, determine the general properties that render it large or small, and calculate its implications for economic innovation. We show that changes to the underlying characteristics of search problems often yield surprising results. For example, more lucrative search rewards can actually reduce the value of mass information, while the possibility of better information often suppresses innovation.

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1 Introduction

What would Thomas Edison have paid for access to Google? His canonical search for a commercially-viable light bulb filament illustrates a broad class of economic decision problems that take the following form. A portfolio of research "leads" is to be sequentially tested for one or more successes. Because each test entails a cost, the leads are ordered according to prior information about their likelihood of success. At some cost, more information is available *en masse* — for example, in the form of an Internet search engine or scientific framework — which informs the researcher about the quality of all leads. Naturally, this will facilitate a more efficient search. The focus of this paper is on the *ex-ante* value of such mass information and on the general implications of search engines for economic innovation and discovery.

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The general characterization of research and development (R&D) as a search problem applies broadly; from economic decisions of grave social importance to the everyday decisions of individual agents. In the case of the incandescent light bulb, Edison's team waded through thousands of research leads[1] before encountering a material, carbonized bamboo thread, that met the requisite criteria for brightness and burn duration. Contemporary examples include searching for a job or employees, a spouse, patentable pharmaceutical compounds in nature, dirty bombs on ships, and even a proof to Fermat's Last Theorem. A common feature of these diverse examples is that they all entail uncertain, sequential, and costly search over a collection of research leads. These features put a premium on search efficiency, and that fact gives rise to our inquiry.

Suppose, for instance, that Edison could have purchased a *search engine* or *scientific database* that contained additional information on the physical and chemical properties of all potential filament materials identified by his team. Such information would clearly have helped to direct their initial searches toward the most promising research leads. How much money would Edison have been willing to pay for access to this type of mass information? Moreover, how would his willingness to pay have been affected by changes to the underlying costs and rewards of developing a light bulb? Our goal is to provide a general framework for answering these types of questions. In so doing, we aim to isolate the factors that make mass information valuable across a broad array of search problems, and determine its consequences for innovation and discovery within the economy.

We proceed by developing a Bayesian decision model of information gathering that informs a generic search process. Analysis of our model then permits us to derive general insights and make three distinct contributions to the literature. First, we derive a simple rule-of-thumb upper bound on the value of mass information in search. In other words, what is the most a researcher would be will to pay for access to a search engine? Second, we identify an important and pervasive tension within the determinants of the value of mass information. This tension drives useful and counterintuitive results that hinge on the underlying characteristics of the search problem. For example, search problems characterized by more lucrative prizes (e.g. the patent on a commercially viable light bulb) often yield smaller value of mass information. Third, we derive the implications of information gathering on economic innovation and show, among other things, that a better search engine can actually lead to less innovation.

1.1 An example

Our basic story can be encapsulated in a stylized example. A farmer wants to sink a water well on his property, but, prior to drilling, he cannot be sure that he will hit water. He initially considers n well locations ("research leads"). Drilling each well costs c (say \$40,000), while he stands to obtain revenue R (say \$200,000) in the event that he strikes water. He requires just one working well and so will cease drilling upon the first success. Based only on prior information, the farmer believes that some locations are more likely to produce a success than others. Because search is costly ($c > 0$), it is optimal for him to search the most promising lead first, the second most promising lead second, and so forth. He therefore searches the leads in descending order of their prior probability of success: $p_{(1)} > p_{(2)} > \cdots > p_{(n)}$, where the subscript refers to the search position. The expected value of such a search conducted in this manner is straightforward to compute.[2]

[1] Including materials as diverse as platinum, paper, vegetable matter and beard hair.
[2] A formal model is derived below. Briefly, the expected value of this search is: $\sum_{i=1}^{N} \Pi_{j=1}^{i-1}(1 - p_{(j)})(p_{(i)}R - c)$.

At some cost z, however, the farmer can acquire a new information set that will allow him to update his beliefs about the potential of all well sites. For example, he could hire a geologist to map his property or he could subscribe to a database that links the probability of well success to observable land features. We loosely refer to this new information set as a *search engine*. Our focus is on the *ex-ante* value of such mass information sources. What is the maximum amount the farmer should rationally pay for access to the geologist's report or database, prior to his knowing the realized information that the new information will provide? Is the amount large or small? On what features of the problem does it depend? And, how might the information affect his choice of drilling order, or the list of potential well sites that he considers viable in the first place? We develop a Bayesian model of mass information gathering to answer these questions as they apply to search problems. Several general principles obtain that can help guide allocations of R&D budgets, as well as provide empirically-interesting insights about the role of search engines in driving economic innovation and discovery.

One of the most salient general principles that emerges is the following. Suppose that an external economic shock raises the value of a working well, R, leaving other features of the problem unchanged. Does this increase in R raise or lower the value of information? Casual intuition might suggest the former, i.e. increasing the value of a working well would make the farmer more willing to pay for the geologist's report. We show that under very reasonable conditions the opposite is often true: Increasing the value of a working water well can actually make the farmer less willing to pay for the geologist's report. A primary mechanism underlying this result is the fact that a higher R raises the economic viability of *all* well sites. This means that the geologist's report may be inconsequential to the final outcome; particularly when R is very high. In a sense, the farmer can afford to be less judicious in his search efforts because most (or even all) well sites will be made worthy of search irrespective of what the geologist might find.

This and other surprising results derive from an interplay between what we refer to as the *intensive* and *extensive* margins of search. Put simply, parameters such as R and c affect the value of information in two ways. Most intuitively, they can affect the value of queuing efficiency that comes with better identifying the most promising leads from a fixed collection (the intensive margin). But they also affect the pool of leads that is worthy of search in the first place (the extensive margin). Under certain conditions, these effects work in tandem. However, under other, often more reasonable conditions, these effects work against each other, yielding new economic insights.

We devote the remainder of this paper to generalizing the problem, solving for the value of information in search and highlighting the most potent results. Following a brief discussion of where this paper fits within the literature, we introduce our theoretical framework in section 2. We then use this framework to derive, analyze, and interpret a variety of theoretical results. We validate our theoretical results with a simulation exercise in section 3, before concluding in section 4.

1.2 The literature on search efficiency and R&D

A concise and important early literature examines optimal search of a collection of leads given fixed and exogenous information about the quality of those leads (Granot and Zuckerman 1991; Gallini and Kotowitz 1985; Roberts and Weitzman 1981; Lucas 1971). Our analysis differs in that we allow for endogenous investment in information about lead quality. We focus on the value of such an investment and on the implications for innovation of this opportunity for information-gathering.

A related literature examines the value of *individual* research leads within a collection. For example, Costello and Ward (2006), Rausser and Small (2000), and Simpson et al. (1996) examine a model of bioprospecting, the search for patentable pharmaceuticals in nature. The key policy question there is whether private sector incentives are sufficient to protect *any given* hectare of land, thus protecting the genetic material it contains. If research leads are of homogeneous quality, the value of any given hectare of land is small, but if leads are of heterogeneous quality, the best leads will be more valuable. Applying our mass information framework to the bioprospecting problem would allow us to examine, for example, the value of indigenous knowledge or biological assays that focused the search on the most promising leads.

As a basis for analysis we consider a version of the canonical "Pandora's Box" search problem solved by Weitzman (1979), which is consistent with other approaches taken in the literature (see, e.g. Roberts and Weitzman (1981), Fudenberg et al. (1983), Gallini and Kotowitz (1985)).[3] For example, Fudenberg et al. (1983) model two stages of R&D: a preliminary invention phase followed by the development phase. Gallini and Kotowitz (1985) similarly distinguish between the basic research and development phases of the R&D pipeline. The purpose of the basic research step is to reduce the number of research avenues so that in the development phase research can focus in on the few most promising projects. More recently, Olszewski and Weber (2015) generalize Weitzman's model so that it accounts for utility from all discoveries and not just the best "box". While the search literature to date has focused on deriving optimal search strategies, we ask: What is the *value* of an efficient search strategy? If R&D projects could benefit nearly as much from, say, random search through a collection of research leads, spending resources on fine-tuning a search strategy might be inefficient.

Our Bayesian setup is holistic because it nests the entire decision problem into a single framework, without needing to separate it into multiple phases of R&D. The *ex-ante* value of a collection of research leads can then be calculated under any information set, and the efficient Bayesian researcher simultaneously endogenizes both the decision about which leads are worthy of search (the extensive margin) *and* the decision about the most efficient order in which to search through them (the intensive margin).

2 A Collection of Leads and the Value of Mass Information

2.1 The value of a collection under prior knowledge

A Bayesian researcher has a set of n statistically independent research leads to be sequentially searched for a success. For expositional clarity, we consider only cases where search terminates upon the first success, in which case revenue R obtains. However, generalizations to multiple successes are straightforward. Testing a lead for a success costs c and the test is without error.

[3] We analyze the problem from the perspective of a single researcher or firm attempting to identify successes among a pool of research leads. This is distinct from the strategic, or "patent race" literature in which multiple firms are competing to achieve the same success (see, e.g. Reinganum (1982), Fudenberg et al. (1983), and Gallini and Kotowitz (1985)).

The researcher's subjective beliefs are captured by her assessment of the prior probabilities of success. For lead i, $p_i^0 = \mathsf{E}_{S_i}(S_i \,|\, I^0)$ where I^0 is the prior information set, S_i is a Boolean $(0, 1)$ success variable for that lead, and the probability p_i^0 is the conditional expectation of success. It will become useful to distinguish between a particular lead (e.g. lead 7 with prior probability p_7^0) and a lead's rank order in the set. Following standard notation in the field of *order statistics*, let $p_{(k)}^0$ be the k^{th} largest value in the set of lead probabilities $\{p^0\}$ under information set I^0. Thus, under I^0, $p_{(1)}^0 \geq p_{(2)}^0 \geq \ldots \geq p_{(n)}^0$.

Because search is costly, the researcher desires an early success. It is straightforward to show that the optimal strategy under I^0 is thus to search leads sequentially in rank order. Leads with probabilities below c/R will be unprofitable to search in expectation, so search terminates either upon the first success or at the final lead with probability that exceeds c/R, whichever comes first. The probability of failing on all leads tested before the ith is $a_i^0 = \prod_{j=1}^{i-1}(1 - p_{(j)}^0)$, with $a_1^0 = 1$.

The expected value of the set of leads $\{p^0\}$ is then

$$V(\{p^0\}) = \sum_{i=1}^{n} a_i^0 (p_{(i)}^0 R - c)\delta_i, \tag{1}$$

where δ_i is a Boolean term that takes value 1 if the corresponding lead is worthy of search ($p_{(i)}^0 > c/R$) and value 0 otherwise ($p_{(i)}^0 \leq c/R$).[4] The value of the collection of leads under information set I^0 can be rewritten in a form with a natural interpretation.

$$V(\{p^0\}) = R\, \overbrace{\left(1 - \prod_{i=1}^{n}(1 - p_{(i)}^0 \delta_i)\right)}^{\text{PS}} - c\, \overbrace{\sum_{i=1}^{n} a_i^0 \delta_i}^{\text{ES}}\,. \tag{2}$$

Here, the coefficient on R is the overall probability of success (PS), and the coefficient on c is the expected number of searches (ES).

2.2 Bayesian updating with a search engine

We now investigate the possibility of acquiring additional information with which the researcher updates her assessment of all lead probabilities. This information could take many forms, and we do not require any structure on the information gathering process beyond our assumption that the information arrives as a set. Convenient mental models include an Internet search engine such as Google, or a scientific framework that provides mass information on the entire set of leads. Note that this implied information set is quite general. For example, it could provide equally useful updates on all leads, or the quality of information could vary across leads. Whatever form it takes, the new information set is denoted I^1, which supplements the initial information I^0. The researcher then updates her probability assessment over lead i as $p_i^{01} = \mathsf{E}_{S_i}(S_i \,|\, I^0 \cup I^1)$.

[4]When $p_{(i)}^0 \geq c/R\ \forall i$, Equation 1 is equivalent to the value of a collection in Rausser and Small (2000). When $p_{(i)}^0 = p \geq c/R\ \forall i$ Equation 1 reduces to the value of a collection in Polasky and Solow (1995, i.e. when $c = 0$), Simpson et al. (1996), and Costello and Ward (2006).

We again adopt the ordered set notation, $p^{01}_{(1)} \geq p^{01}_{(2)} \geq \ldots \geq p^{01}_{(n)}$. Importantly, because a particular lead's assessment changes with information (i.e. $p^0_i \neq p^{01}_i$) the researcher may reorder the leads after I^1 has arrived. Thus, the lead that is ranked k^{th} under prior information ($p^0_{(k)}$) may differ from the lead that is ranked k^{th} under updated information ($p^{01}_{(k)}$). Upon receipt of the new information and subsequent updating, the revised expected value of the collection with the updated information is simply $V(\{p^{01}\})$ applying equation (1).

What is the value of this mass information set I^1? The *ex-post* value is simply the difference: $V(\{p^{01}\}) - V(\{p^0\})$. But this formula is of no real use to the researcher *ex-ante* when she must decide whether to acquire the new information set before knowing its actual contents. To form an *ex-ante* expectation over the value of information, the researcher must instead account for her present uncertainty by treating I^1 as random with some specified probability distribution. Since $p^{01}_i = \mathsf{E}_{S_i}(S_i \mid I^0 \cup I^1)$, the distribution of I^1 in turn induces a probability distribution for each lead over $\{p^{01}\}$ by standard Bayesian updating. The *ex-ante* prediction of this distribution is called a predictive posterior or, simply, a *preposterior* distribution.

2.3 Preposterior distributions and the margins of search

Preposterior distributions characterize the ways in which new information might conceivably affect our beliefs. They can therefore be thought of as a meta-distribution of all *possible* posterior outcomes, given different realizations of an information set. For current information set I^0, let $g_j(p^{01}_j)$ be the preposterior distribution of the success probability for lead j. This distribution has some concrete properties. First, to ensure Bayesian consistency, the mean of the preposterior distribution must equal the mean of the prior distribution. If not, the researcher would be able to change her prior belief without obtaining additional information. The result is highlighted below:

Lemma 1: The ex-ante expectation of the posterior belief about success probability for any lead j equals the prior belief about success for that lead.

Proof. $\mathsf{E}_{I^1}(p^{01}_j) = \mathsf{E}_{I^1}(\mathsf{E}_{S_j}(S_j \mid I^0 \cup I^1)) = \mathsf{E}_{S_j}(S_j \mid I^0) = p^0_j$. The first and last equalities are definitions. The central equality is the law of iterated expectations. □

Lemma 1 shows that the mean of $g_j(p^{01}_j)$ must equal p^0_j itself. The spread of the preposterior distribution for any lead will thus reflect our expectations about the quality of information — relative to the strength of our priors — on that lead. Loosely speaking, if the information is expected to be superb, the distribution of posterior means will be highly spread out because information is expected to substantially alter the researcher's beliefs. On the other hand, if the information is unlikely to alter the researcher's beliefs about lead quality, then the distribution $g(\cdot)$ will be tightly concentrated around the prior belief, so that $p^{01} \simeq p^0$. This result is formalized by the observation below:

Lemma 2: More information about a lead induces a mean-preserving spread over its preposterior distribution.

Proof. Trivially, $p_j^{01} = p_j^0 + (p_j^{01} - p_j^0)$. By Lemma 1, the term $(p_j^{01} - p_j^0)$ is mean zero. Similarly, $(p_j^{01} - p_j^0)$ is unconditionally orthogonal to p_j^0. $\mathsf{E}_{I^0,I^1}\left(p^0(p^{01} - p^0)\right) = \mathsf{E}_{I^0}\left(p^0 \mathsf{E}_{I^1}(p^{01} - p^0)\right) = 0$. □

Lemma 2 uses the standard definition of mean-preserving spread provided by Rothschild and Stiglitz (1970). As a corollary, it follows further from their analysis of mean-preserving spreads that a preposterior based on more information second-order *stochastically dominates* a preposterior based on less information.

Lemmas 1 and 2 together yield some important and generalizable insights for our framework. For example, better information causes the preposterior to become more dispersed. This again reflects the fact that good information has the potential to substantially alter beliefs, whereas poor information does not. Similarly, the spread in preposteriors is a weighted mix of prior beliefs and new information. Particularly strong priors will tend to dominate the final outcome in a way that obviates the potential impact of additional information. The key insight for our framework, however, is the fact that mass information gathering in search operates through two distinct channels. First and most intuitively, information can cause the researcher to re-order the collection of leads. This results in an efficiency gain because higher probability leads are ordered earlier in the queue. We refer to this effect as the *intensive margin* of search. The second, more nuanced effect is that information can change the set of leads that are worthy of search. For example, some lead k with prior probability $p_k^0 > c/R$ could have posterior probability $p_k^{01} < c/R$, in which case the researcher will rationally drop it from the collection upon receipt of the new information set. We refer to this effect as the *extensive margin* of search. We next turn to disentangling these two effects before combining them into a holistic decision framework. Before doing so, however, it is useful to clarify a key aspect of the preposterior and information gathering process.

When we say that a researcher can purchase additional information about a set of leads, we do not imply that this information always serves as a perfect substitute for actual testing. Only testing a lead — at some cost — can truly determine whether it yields a success or failure. Information, on the other hand, simply tells us more about the true probability distribution of the lead.[5] Even very good information about the properties of carbonized bamboo thread would not have guaranteed the success of Edison's light bulb before his team had physically tested it. The importance of this distinction will become apparent as we develop our framework in the subsequent sections.

2.3.1 The *intensive margin*

The most obvious dividend from mass information is that it enables a more efficient search ordering. To explore this intensive margin of search, consider an $n = 2$ lead case with overlapping preposterior distributions, but where there is no mass from either distribution below c/R: $g_i(p_i^{01}) = 0$ for $p_i^{01} \leq c/R$, $i = 1, 2$. By Lemma 1 the prior means must satisfy: $p_1^0, p_2^0 > c/R$, so both leads are initially worthy of search. Further there is no possibility that either lead will be dropped from the collection with information, because the preposteriors have no mass below c/R. Thus, there is no extensive margin of search. The intensive margin is present, however, because the supports overlap and so the rank order of leads may change with additional information I^1. This reordering has value because it saves searches in expectation.

[5] Only in the extreme case (where the preposterior is a two-point distribution on (0,1)) does information replace testing. See Section 2.3.2.

Consider the following example: You seek an ideal life partner from a collection of 2 individuals whose qualities are uncertain. You assign the following prior probabilities of success: $p_1^0 = .7$, $p_2^0 = .5$. The cost of courtship is c =\$1,000 and this will reveal with certainty whether a candidate (i.e. lead) is a success or a failure. The gross payoff upon success is R =\$10,000. Note that both candidates are worthy of search under your prior belief probabilities: $p_1^0 > p_2^0 > c/R$. Consider then what would happen if you have the chance to acquire additional information about these two candidates, e.g. by obtaining a compatibility profile from a dating service. The information acquired will not be perfect, but will allow you, in expectation, to update your probability assessments. To illustrate the point, suppose that both preposterior distributions are two-point distributions:

$$p_1^{01} = \begin{cases} .5 \text{ with probability } 1/2 \\ .9 \text{ with probability } 1/2 \end{cases} \tag{3}$$

$$p_2^{01} = \begin{cases} .3 \text{ with probability } 1/2 \\ .7 \text{ with probability } 1/2 \end{cases} \tag{4}$$

Note that the extensive margin does not exist under these preposterior distributions. No information from the dating service would cause you to drop a candidate from the worthy collection. However, you may decide to re-order your search. Under prior information only, efficiency demands that you search candidate 1 first and then, conditional upon failure, candidate 2. As per equation (2), this in turn implies that the collection will yield an initial expected payoff of \$7,200. In contrast, the expected value of the collection with the dating service is \$7,250. The *ex-ante* expected value of information is thus \$50 and, moreover, accrues entirely to the intensive margin of search. Subscribing to the dating service would be a sound investment if it costs less than this.

This simple example highlights an important principle underlying the intensive margin of search. Information cannot change the collection of worthy leads if the preposterior does not span the truncation point, c/R. However, it may well change the order in which the leads are searched. More formally: In the n-lead search problem with overlapping preposterior distributions, if the supports do not span c/R, then the entire value of information accrues to the intensive margin of search.[6]

2.3.2 The *extensive margin*

The extensive margin of search — which leads are worthy of search — can be isolated in a similar manner. Consider the single lead problem of whether to go to a movie. The cost of the movie is c = \$10 and the revenue upon success (i.e. the gross payoff from watching a good movie) is R = \$20. *Ex-ante* the quality of the movie (whether it will produce a "success") is uncertain but at some cost you can acquire additional information, e.g. by reading a movie review. In this simple case, the intensive margin is absent; there is no possibility of re-ordering the leads because only one lead exists. However, information can still have value owing to the extensive margin. You expect that reading the review will change your assessment of the movie's quality (success probability),

[6]This observation parallels "Cromwell's Rule" (Jackman 2009), which declares that a posterior distribution cannot allocate any density to values that are explicitly excluded by the prior. As we will soon see, a similar logic holds with respect to non-overlapping preposterior densities and the extensive margin of search.

although by Lemma 1 you do not know whether it will increase or decrease. If your assessment changes significantly, it may change your decision about whether to attend the movie at all, in which case the information will have provided value. If the review is from a highly trustworthy source — say, a movie critic with preferences identical to yours — then the preposterior is a two point distribution, as follows:

$$p^{01} = \begin{cases} 0 \text{ with probability } 1 - p^0 \\ 1 \text{ with probability } p^0 \end{cases} \tag{5}$$

A less informative reviewer would induce a less spread out preposterior.

Suppose your prior assessment is $p^0 = .6$, so that you believe there is a 60% chance the movie will be a "success" and a 40% chance the movie will be a "failure". Without additional information, you would attend the movie because attendance has a positive expected value: $p^0 R - c = \$2$. *Ex-ante*, the expected value of this single-lead collection with the movie review is $p^0(R-c)$ =\$6. How much would you pay for access to the trustworthy review? Simply the difference: \$4.[7] Importantly, the extensive margin can also have the opposite effect. If your prior belief was $p^0 = .3$ (rather than .6), then without the information, you would elect to skip the movie, but upon reading the review you may decide to go. In this case, the value of the trustworthy review is \$3.

This simple example highlights the general role of the extensive margin in search. Information can either expand or contract the collection of worthy leads. Note further that if two preposterior distributions do not overlap, the left-most distribution will always have the lower final realization. There is no chance that information will reverse the relative ranking of success probabilities for these two leads. More formally: In the n-lead search problem with non-overlapping preposterior distributions, the entire value of information accrues to the extensive margin of search.

2.4 A general formula for the value of mass information in search

We have argued that search engines and other providers of mass information can yield two distinct dividends in search, which we have characterized as the extensive and intensive margins. Thus far we have artificially separated these two effects to develop an intuition around their individual roles. However, for many search problems the two are inextricably linked and can even have counteracting consequences. The remainder of this paper is devoted to combining the effects in a coherent Bayesian model and deriving general principles about the value of information in search.

Consider the general two-lead ($n = 2$) case. The *ex-ante* value of this collection is derived by taking the expectation of equation (1) over the random variables p_1^{01} and p_2^{01}, for which we assign integration dummies $\tilde{p}_1$ and $\tilde{p}_2$, respectively. The selected search queue depends on whether $p_1^{01} \lessgtr p_2^{01}$, so we divide the expectation integral into two regions, and the *ex-ante* expected value of

[7] From the preposterior payoffs: $\left(1 - p^0\right)(-x) + p^0 (R - c - x)) = \ldots = \$6 - x$. With a prior expected value of $x = \$2$, we obtain \$4.

the collection is:

$$\mathsf{E}_{I^1} V(\{p_1^{01}, p_2^{01}\}) = \int_{c/R}^{1} \left[\int_{c/R}^{\tilde{p}_1} ((\tilde{p}_1 R - c) + (1 - \tilde{p}_1)(\tilde{p}_2 R - c)) g_2(\tilde{p}_2) \right] g_1(\tilde{p}_1) + \int_{c/R}^{1} \left[\int_{c/R}^{\tilde{p}_2} ((\tilde{p}_2 R - c) + (1 - \tilde{p}_2)(\tilde{p}_1 R - c)) g_1(\tilde{p}_1) \right] g_2(\tilde{p}_2) \tag{6}$$

The first term on the right-hand side is the *ex-ante* expected value of the collection when $p_1^{01} > p_2^{01}$, multiplied by the probability of this event occurring. The second term reverses the inequality. It is convenient to re-write this expression in terms of the order statistics $p_{(1)}^{01}$ and $p_{(2)}^{01}$, for which we use integration dummies $\tilde{p}_{(1)}$ and $\tilde{p}_{(2)}$, where $\tilde{p}_{(1)} = \tilde{p}_1$ in the first integral and $\tilde{p}_{(1)} = \tilde{p}_2$ in the second.

$$\mathsf{E}_{I^1} V(\{p_1^{01}, p_2^{01}\}) = \int_{c/R}^{1} \left[\int_{c/R}^{\tilde{p}_{(1)}} ((\tilde{p}_{(1)} R - c) + (1 - \tilde{p}_{(1)})(\tilde{p}_{(2)} R - c)) g_2(\tilde{p}_{(2)}) \right] g_1(\tilde{p}_{(1)}) + \int_{c/R}^{1} \left[\int_{c/R}^{\tilde{p}_{(1)}} ((\tilde{p}_{(1)} R - c) + (1 - \tilde{p}_{(1)})(\tilde{p}_{(1)} R - c)) g_1(\tilde{p}_{(1)}) \right] g_2(\tilde{p}_{(1)}) \tag{7}$$

Which reduces to

$$\mathsf{E}_{I^1} V(\{p_1^{01}, p_2^{01}\}) = \int_{c/R}^{1} \int_{c/R}^{\tilde{p}_{(1)}} ((\tilde{p}_{(1)} R - c) + (1 - \tilde{p}_{(1)})(\tilde{p}_{(2)} R - c)) \left[g_2(\tilde{p}_{(2)}) g_1(\tilde{p}_{(1)}) + g_1(\tilde{p}_{(1)}) g_2(\tilde{p}_{(1)}) \right]. \tag{8}$$

Again invoking order statistics, the general formula for the *ex-ante* expected value of a collection of n leads with (forthcoming) information I^1 is:

$$\mathsf{E}_{I^1} V(\{p^{01}\}) = \sum_{\sigma(r,s,\ldots,t)} \int_{c/R}^{1} \int_{c/R}^{\tilde{p}_{(1)}} \cdots \int_{c/R}^{\tilde{p}_{(n-1)}} \sum_{i=1}^{n} b_i (\tilde{p}_{(i)} R - c) g_r(\tilde{p}_{(1)}) g_s(\tilde{p}_{(2)}) \ldots g_t(\tilde{p}_{(n)}). \tag{9}$$

where $b_j = \prod_{j=1}^{i-1} (1 - \tilde{p}_{(j)})$, with $b_1 = 1$. The first summation is over all permutations $\sigma(r, s, \ldots, t)$ of the indices of the preposterior distributions. This permutation reflects, as above, that the highest realized probability $p_{(1)}^{01}$ does not necessarily arise from the lead with the highest prior probability $p_{(1)}^{0}$.

Equation (9) provides a general formula for the *ex-ante* value of a collection of research leads when mass information, I^1, becomes available. The value of a search engine or scientific framework, here denoted $\Theta(I^1)$, is then the difference between the expected value of the collection with and without the mass information that it provides:

$$\Theta(I^1) = \mathsf{E}_{I^1} V(\{p^{01}\}) - V(\{p^0\}) \tag{10}$$

Proposition 1: $\Theta(I^1)$ is (weakly) increasing in a mean-preserving spread over any of the preposterior distributions.

Proof. The proposition follows from two observations. First, the expected value of search is individually (though not jointly) convex in the success probability of each lead. This can be seen by inspecting equation (1) for the value of search over the prior set, $\{p^0\}$. This function is individually piecewise linear in p_i^0, so that the derivative is locally constant.[8] For lower p_i^0's the derivative is smaller, because of the summation on the cost term. As the value of search is individually convex for known p_i^0's, the *expected* value of search over unknown p_i^0's must also be individually convex, since the expectation is a weighted average of convex functions. Second, Rothschild and Stiglitz (1970) demonstrate that the expectation of any individually convex function of a random variable p cannot decrease from a mean-preserving spread of the distribution of p. □

Proposition 1 accords with intuition; especially in light of Lemma 2. A more dispersed preposterior distribution means that the researcher's subjective beliefs are more likely to change after acquiring I^1. Taken together, Proposition 1 and Lemma 2 also imply that the expected value of additional information in search is non-negative. Indeed, it is easily verified that the expected value of information is positive if and only if the preposterior distributions overlap above c/R. Additional information cannot harm a searcher, since it does not change her feasible search set. Moreover, since information must be adding value if it is going to change her actions, this implies that the preposteriors must be overlapping.

2.5 Determinants of the value of mass information

Thus far our focus has been on how the value of mass information is affected by changes in informational quality and quantity.

However, our larger goal is to characterise the conditions under which search engines and other providers of mass information become relevant as meaningful drivers of economic activity and decision-making. For example, would Thomas Edison have been more or less willing to pay for Google if he was guaranteed a longer patent on the incandescent light bulb? Similarly, is there an upper bound on the amount that he would have been willing to pay, and on what does it depend? With these questions in mind, we next investigate the role played by the other economic parameters of our generalized search model in determining the value of mass information, holding constant the information that may be acquired.

2.5.1 Role of the economic parameters: R and c

Proposition 2: $\frac{\partial \Theta(I^1)}{\partial R}\big|_{p_i^0 \neq c/R}$ = The expected change in the (local) probability of success due to acquiring new information.

Proof. Recall that $\Theta(I^1)$ is a continuous, yet non-smooth function: Its derivative with respect to R is not defined at the truncation points where $p_i^0 = c/R$. With that in mind, the proof applies to local regions of the function in between truncation points and is easily seen by separating $\Theta(I^1)$ into its constituent parts of uninformed and informed search. As per equation (2), the value of uniformed search, $V(\{p^0\})$, is linear in R with a coefficient equal to the probability of success. Similarly, as per equation (9), the expected value of informed search, $\mathsf{E}_{I^1} V(\{p^{01}\})$, enters the integrand linearly

[8] Note that while $\Theta(I^1)$ is a continuous function, it is not smooth. The first derivative is noncontinuous because of the truncation points that are introduced when information causes an additional lead to become worthy of search (or, alternatively causes it to dropped from the worthy pool).

multiplied by the probability of success. The total derivative is thus simply the difference between the expected probability of success under informed and uniformed search. □

Proposition 3: $\frac{\partial\Theta(I^1)}{\partial c}|_{p_i^0 \neq c/R}$ = The expected change in the (local) number of searches due to acquiring new information.

Proof. The proof proceeds in an identical manner to the previous proposition. The only difference being that the relevant linear coefficient on c is such that the derivative is equal to the difference in the (expected) number of searches that arise under informed versus uniformed search. □

Proposition 4: $\Theta(I^1)$ has an interior maximum in both R and c.

Proof. Uninformed search has zero value if $c/R \geq p^0_{(1)}$. In such a case, $\Theta(I^1)$ is simply the value of informed search, which must be decreasing in c and increasing in R over this range. In contrast, the full set of leads will be evaluated under uninformed search if $c/R \leq p^0_{(n)}$. Yet, new information may cause some leads to be dropped from the worthy pool if any preposterior distribution has positive mass $< c/R$. The probability of success and the expected number of searches will hence be lower for informed search. Propositions 2 and 3 show that these are equal to the derivatives of informed search with respect to R and c. Therefore, $\Theta(I^1)$ is increasing in c and decreasing in R over this range. □

Proposition 4 implies that information can be increasing or decreasing in both R and c. We shall return to this proposition in Section 3 and demonstrate its economic intuition with the aid of a simulation exercise. As a matter of completeness, it should also be noted that the value of information is homogeneous of degree one in R and c.[9] Intuitively, this property of linear homogeneity must hold, otherwise arbitrarily coincident changes in the parameters — e.g. changing units from dollars to yen — would fundamentally change the analysis.

2.5.2 Maximal value of mass information in search

For an exogenously defined set of prior probabilities, preposterior distributions, and economic parameters, we have derived some general results and comparative statics regarding the value of mass information in search. While these results can provide some guidance for efficient investment in information, we provide here an extremely simple rule-of-thumb for this investment. To do so, we first derive the collection of prior probabilities and associated preposterior probabilities that theoretically maximize the value of information in search. We then derive a closed-form expression for Θ, and interpret it as an upper bound on the value of information in search.

Proposition 5: The maximal value of mass information in search is:

$$\bar{\Theta}(I^1) = Rn(1+n)^{-\frac{n+1}{n}} \tag{11}$$

[9] This can be seen by inspecting $V(\{p^0\})$ and the integrand of $\mathsf{E}_{I^1}V(\{p^{01}\})$ in equation (10). The parameters also enter the limits of integration of $\mathsf{E}_{I^1}V(\{p^{01}\})$, but invariably enter as the fraction, c/R, and so a multiplicative factor is of no consequence.

Proof. By Proposition 1, a mean preserving spread increases the value of information. For any set of prior leads, $\{p^0\}$, the maximal spread preposterior is full knowledge. That is, a two-point distribution with mass p_i^0 at one, and mass $1 - p_i^0$ at zero. For this preposterior, the expected value of informed search is $(1 - \prod(1 - p_i^0))(R - c)$. This follows because the searcher knows before bearing any search costs whether any of the leads is a success. And if so, he will only test a successful lead. The expected value for uninformed search for that set of $p's$ is given by equation (1). It can easily be shown that the derivative of $\Theta(I^1)$ with respect to each p is negative. So, the value of information is maximized at the lowest set of p's worthy of search. That is, it obtains at $p = c/R$ for each p. At this point, uninformed search has no value, so $\Theta(I^1) = (1-(1-c/R)^n)(R-c) = R(1-(1-c/R)^n)(1-c/R)$. Maximizing with respect to c and algebraically simplifying yields the result. □

The bound described in equation (11) requires knowledge of only R and n. The bound is increasing in n; for large n, the maximal value of information approaches R. But for small n, this formula can actually help guide investment. Take $n = 1$ for example, where the bound is simply $R/4$. There, one should never invest more in information acquisition than $1/4$ of the potential return from a successful search.

The general conclusion that one would be willing to pay more for a search engine with a larger pool of leads (n) can be shown to hold more generally. However, one must take care to define which leads are being added to the pool. A fair comparison involves adding leads that are, *ex-ante*, indistinguishable from existing leads. For example, suppose all leads have the same prior probabilities of success and also share the same preposterior distributions. In that case, it is easy to show that $\frac{d\Theta(I^1)}{dn} > 0$.

2.6 Mass information aquistion and innovation

Thus far we have focused on the technical aspects of mass information gathering in search. We can also deploy our framework to conduct a more positive analysis. In particular, we can examine the surprising effects of mass information acquisition on innovation. Note that innovation in our model is a Bernoulli random variable, which is described by the probability of a successful search.[10] Since our formula for $\Theta(I^1)$ already incorporates changes in the probability of success — see also equation (2) — this means that it can be used to determine the effects of new information on innovation. Given this setup, a natural question to ask then is how innovation would respond to an exogenous increase in information (e.g. the introduction of Google, or an improved search algorithm). Casual intuition might suggest that the amount of innovation would necessarily increase after such an information shock. However, this turns out not to be true, as shown below:

Proposition 6: An exogenous increase in the amount of information can either increase or decrease the expected level of innovation.

Proof. The single lead case provides the most straightforward proof by example. If $p^0 > c/R$, then the lead is searched for certain without additional information, but possibly not with it.

[10] For example, innovation in the medical field of oncology would be characterized by the probability of finding a successful cure for cancer. In the case of online dating services like eHarmony and Match.com, the equivalent metric might be the number of successful marriages.

Information could therefore lead to less innovation. If we instead had a situation where $p^0 < c/R$, then the result reverses. Information could cause the lead to be included as a viable search possibility, thus leading to more innovation. □

The basic intuition for Proposition 6 is best understood through the lens of the extensive margin. Information can cause the pool of acceptable leads to either grow or to shrink. These changes could have opposing effects on the probability of success (i.e. innovation). From a societal perspective, we may want researchers to take risks that are not necessarily rational from an individual perspective — insofar as these risks aren't supported by their informed expected utility calculus. This is simply to say that innovation confers positive spillovers to society and the risks to R&D should be distributed appropriately. As a corollary, note too that a change in the price of information would have an analogous effect. For example, suppose that Google decided to charge a subscription fee for its use. Making information more expensive could dissuade a researcher from acquiring it, even if she had intended to previously. As we have just established, this could in turn either increase or decrease the probability of innovation.

While Proposition 6 involves an exogenous increase in information, our model can also account for innovation that arises from *endogenous* information gathering decisions. That is to say, in situations where researchers decide to acquire additional information only if it is economically rational to do so.

Proposition 7: Endogenous innovation is weakly increasing in R.

Proof. An efficient researcher will not acquire information when it is unprofitable, e.g. when information is prohibitively expensive. Consider then a scenario in which a researcher initially finds it profitable to acquire information about a pool of leads. How would innovation respond to an increase in R under this scenario? There are two cases that we need to evaluate.

(i) *The increase in R has no effect on the researcher's decision to acquire information.* In this case, we are comparing expected innovation under two instances of informed search, but with different levels of R. Since an increase in R can never decrease the pool of worthy leads for a given information set, it cannot decrease the level of innovation.

(ii) *The increase in R causes the researcher to stop acquiring information.* In this case, we are comparing expected innovation under informed search with a given R to innovation under uninformed search with a higher R. For local increases in R that do not span separate truncation points, information gathering can only be made unprofitable if $\partial\Theta(I^1)/\partial R < 0$. However, as per Proposition 2, this would also imply that the switch to uninformed search (prompted by the higher R) would yield more innovation. For larger increases in R that do span truncation points, additional leads will be added to the worthy pool in a way that also guarantees more innovation irrespective of whether information is acquired or not. To see this, note that a larger increase in R can itself be decomposed into two movements. The first is a jump across truncation points that preserves the existing decision about the need to acquire information. (For example, an increase in R that is calibrated to leave $\Theta(I^1)$ exactly unchanged.) Since this jump involves evaluating additional leads under the same information set, it cannot decrease the level of innovation. The second movement is a local adjustment that brings R to its final position within the new truncation span, below the required threshold for acquiring information. As already described above, however, this latter movement implies an expected level of endogenous innovation that is higher for uninformed search (again, compare Proposition 2).

Taken together, these two cases show that the overall level of innovation is weakly increasing in R irrespective of whether it causes the researcher to stop acquiring information. □

While the above proof is specific to a scenario in which the researcher initially chooses to acquire information, it is straightforward to apply identical logic to an alternative scenario in which she initially favors uninformed search. Again, the result is that an increase in R either leaves her decision about information gathering unchanged, or induces endogenous information gathering in a way that cannot decrease innovation. Similarly, note that an increase in R always leads to more (or at worst unchanged) innovation in our model, regardless of its impact on our decision to acquire information. Offering more patent protection on cancer medication, for example, cannot decrease innovation in the field of oncology. But it may lower our willingness to acquire mass information on the pool of potential treatments.

Proposition 8: The amount of endogenous innovation is weakly decreasing in c.

Proof. The proof proceeds in an identical manner to the previous proposition, except that the mechanism involves a change in the expected number of searches. Intuitively, an increase in the costs of testing individual leads will lead to less research activity, thus lowering the overall level of endogenous innovation. We confirm this property with the aid of simulations in the next section. □

Taken together, Propositions 6-8 have important implications for how society funds innovation, as well as how applied researchers measure the empirical rates of innovation over time. For example, rather than providing broad support for basic research (increasing background knowledge and lowering the cost of acquiring information), our results may suggest that organizations such as the National Science Foundation would stimulate innovation in the economy more effectively by funding research prizes, expanding patent protection, or even increasing the grant endowments of a few select research groups. Our results also suggest that patent data will not necessarily provide a reliable proxy for empirical measures of innovation over time. For example, a surge in patent filings may simply reflect an exogenous increase in background scientific knowledge or information provision. Such drivers need not corresponded to an environment of increased innovation. Indeed, they could actively induce a *decrease* in innovation as per Proposition 6. On the other hand, if patent filing is positively correlated with search rewards, then patent data would provide an accurate measure of innovation as per Proposition 7. The larger point that we wish to make here is that there are various drivers of patent filings, each of which may be more or less correlated with true levels of innovation. Understanding which of these drivers dominates the final outcome is crucial to patent data's empirical justification as a measure of innovation.

3 Simulation

To help convey the underlying economic intuition of the theoretical results from the previous sections, we return to our initial example: A farmer's search for a suitable water well. In particular,

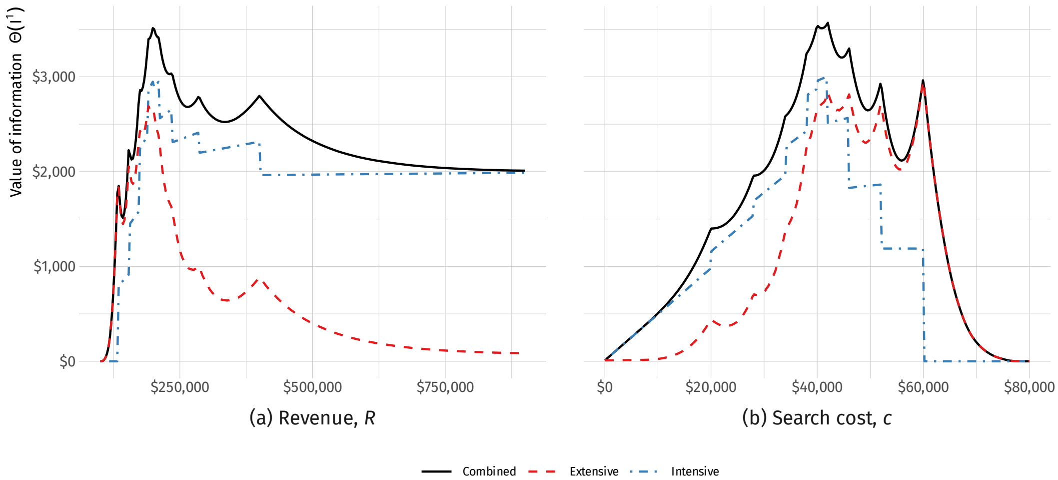

Figure 1: Value of information gathering in search as function of parameters.

Note: Kinks represent points of truncation for the prior-information search. These are points where $p_i^0 = c/R$.

we simulate the search problem with $n = 9$ possible well sites, $R = \$200,000$, and $c = \$40,000$.[11] Under these parameters, the value of information in search is about $\Theta(I^1) = \$3,600$, or roughly 10% of the expected search value. The geologist's report should be acquired if it costs less than this. How, then, do changes in the parameters affect the farmer's willingness to acquire information? The simulation results are depicted in Figure 1, where the combined value of information is presented alongside the decomposed contributions of the intensive and extensive margins on their own.

Consider first a change in R, as shown in Figure 1(a). When R is sufficiently small, sinking a successful well yields insufficient gains irrespective of the farmer's knowledge and so the value of new information is zero. However, as R begins to increase, the farmer expects that new information will reveal some well sites ("leads") to be worthy of search. This includes well sites that he would not have considered based on his prior knowledge alone; highlighting the role of the extensive margin. The value of information is therefore initially increasing in R. At the same time, there exists a sufficiently high value of R for which the farmer searches *all* leads given his prior knowledge. Yet, he also expects that new information will enable him to effectively discard some of these leads. The pool of leads is therefore smaller at this higher level of R with information than without it. This effect saturates at extremely large values of R. All n leads are now searched regardless, so that the only value of additional information is facilitating a more efficient search ordering. The value of information beyond this point is entirely captured by the intensive margin. A similar effect is realized for increases in c, as per Figure 1(b). The only real difference is that when c gets sufficiently large, no leads are worthy of search and the value of information collapses to zero.

Figure 1 not only depicts the relative contributions of the intensive and extensive margins to the overall value of information in search. It also helps visualize the costs that arise from ignoring either one of the individual margins. When R is "large" relative to c, then the intensive margin

[11] We use prior probabilities $\{.10, .14, .17, .19, .20, .21, .23, .26, .30\}$ for the different well sites, while the preposteriors are assumed to follow beta distributions with a constant variance .001. Simulation code is available upon request.

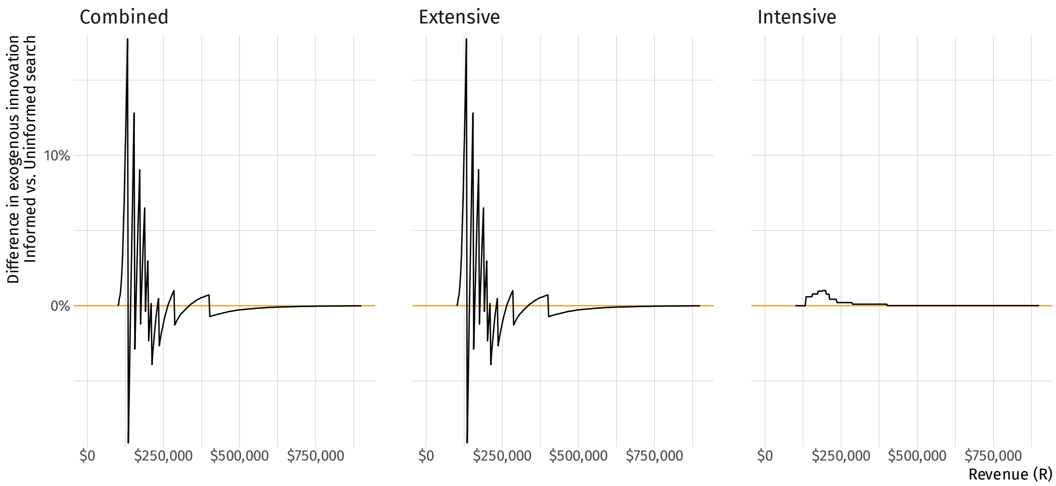

Figure 2: Exogenous innovation as function of information.

will do a better job of approximating the combined, true value of information than the extensive margin (and vice versa). Similarly, while the individual margins peak around the same parameter values as the combined series, both will tend to underestimate the true value of information. It should also be noted that the value of information is not in general additive. That is to say, the combined value of information does not equal the summed values of the intensive and extensive margins.

We can also analyze the effect of changing parameter values on innovation.[12] Figure 2 shows how innovation responds to an exogenous shock in information by comparing the differences between informed and uninformed search. The figure not only confirms the fact that an exogenous information shock can either increase or decrease the level of innovation (Proposition 6), but also illustrates that the extensive margin is the primary channel for this phenomenon. For example, innovation is initially higher for informed search than uninformed search at very low levels of R. This is because even the farmer's best lead isn't searched under prior knowledge when R is very small, i.e. $p^0_{(1)} < c/R$. However, new information sometimes reveals that this lead (and possibly others in the search queue) are actually worthy of search, hence yielding higher innovation in expectation. The reverse is true for very high levels of R, where all leads are searched on the basis of prior knowledge alone, but not necessarily with additional information.

We can similarly inspect the response of *endogenous* innovation — where the researcher optimally decides whether to acquire information — to changing search parameter values. In Figure 3, we assume that the geologist's report costs $\$2,500$ to acquire. The underlying simulations confirm our theoretical predictions that endogenous innovation is weakly increasing in R and weakly decreasing in c (Propositions 7 and 8).

[12]While the role of innovation is perhaps hard to interpret in the context of a farmer searching for a well, the general principles that we established in the preceding sections would all still apply. Innovation corresponds to the probability of success.

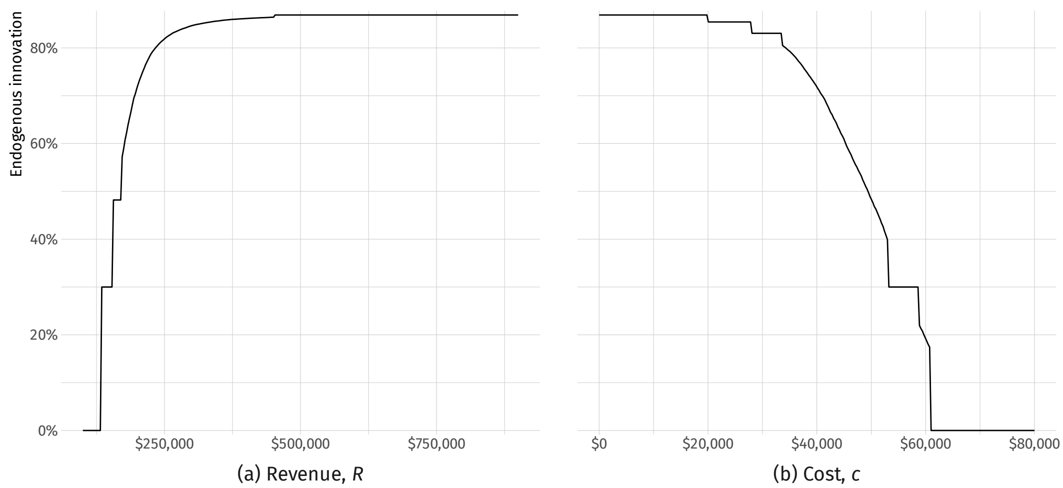

Figure 3: Endogenous innovation as function of search parameters.

Note: The cost of information is assumed to be z = $2,500. Sharp transitions denote truncation points in the parameter space where uninformed search is optimal, i.e. $z > \Theta(I^1)$. Smooth transitions denote truncation points in the parameter space where informed search is optimal, i.e. $z < \Theta(I^1)$. Compare with Figure 1.

4 Discussion

The main purpose of research is to resolve uncertainty and inform better decision-making. Whether one seeks a new medicine, a marketable invention, or an elegant proof to a theorem, a collection of leads could be searched on the basis of only prior information. Often the researcher has the opportunity to better organize her search by acquiring additional information about a set of research leads *en masse*, for example with an search engine or scientific framework. We have developed a framework within which to analyse the value of such mass information gathering in search.

We have found that for any search problem with these basic characteristics, information yields two important, and sometime counteracting, dividends. The most intuitive dividend from information is that it facilitates a more efficient search order, and thus allows the researcher to obtain success with fewer searches. If each search is costly (c is large) this effect can be large. But there is a less obvious, and perhaps more powerful, dividend from information. Improved information about the research leads' qualities may reveal that some leads which appeared to be economically viable candidates with only prior information ($p_i^0 > c/R$), are no longer viable ($p_i^{01} < c/R$). Thus, information may *reduce*, perhaps to large measure, the collection of viable leads. Eliminating large numbers of unprofitable leads from the collection can have significant value. Roughly speaking, the more likely it is that unprofitable leads will be exposed by information (and thus eliminated), the larger is the value of information. But this effect can also work in the opposite direction. Information can reveal that large numbers of leads once thought to be impotent (on the basis of prior information only), should indeed be searched. This juxtaposition gives rise to several interesting generalities regarding the conditions under which information has high value and its implications for innovation. For example, we have argued that innovation would be more effectively encouraged in the economy by promoting research competitions such as XPRIZE[13] and lowering the costs of testing individual leads. While increased innovation does not necessarily translate to improvements in welfare, our analysis does suggest some fairly concrete mechanisms by which policymakers can direct the two. Moreover, given the positive economic spillovers associated with innovation and R&D, the provision of mass information may provide the key conduit through which society is able to internalize the inherent risks and rewards of these crucial activities.

The analysis presented here is not without its limitations. We have assumed statistical independence among the leads. Solving the optimal sequential search problem with statistical dependence is a difficult task, but once that ordering is obtained, the insights uncovered by this analysis would be qualitatively unchanged. We have also assumed that the "quality" of information (captured by the spread of the preposterior distribution) was exogenously given, and could not be chosen by the researcher. While the exact mechanism of information gathering and updating was not the focus of this paper, one can imagine extensions that examine, for example, the optimal quality of a search engine or dating service. The framework developed here would inform the value of information of different qualities, and those would have to be compared against the costs.

The basic principles analyzed here apply broadly to any research inquiry. Researchers face the two basic questions of *what* leads to search and *how* to order the search queue of the worthy collection. Considering both effects simultaneously uncovers economic insights about the efficiency of information gathering and its dependence on features of a search problem. Among other novel properties, we find that mass information can: (1) suppress innovation, (2) have value that is negatively related to the revenue upon success, and (3) have value that is negatively related to the cost of testing leads.

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[13] `http://www.xprize.org/`

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